# Lecture 7. Estimation of Basic Quantities for Other Sampling Schemes

Mean and Quantiles; Left Truncation; Left Censoring; Right Truncation; Turnbull

## Contents


**1 Estimating the Mean Survival Time** **1**

1.1 Mean as the area under the curve . . . . . 1

1.2 First solution: Efron's completion . . . . . 1

1.3 Second solution: restricted mean . . . . . 1

**2 Quantiles and Their Confidence Intervals** **2**

**3 Right-Censored and Left-Truncated Data** **3**

**4 Left-Censored Data: Reversing the Time Scale** **3**

**5 Right-Truncated Data** **4**

**6 Other Sampling Schemes** **4**


## 1 Estimating the Mean Survival Time

### 1.1 Mean as the area under the curve

Since $\mu = \mathrm{E}(X) = \int_0^\infty S(t)\,\mathrm{d}t$, the natural estimator is $\hat{\mu} = \int_0^\infty \hat{S}(t)\,\mathrm{d}t$. Problem: if the last observation is censored, $\hat{S}(t)$ stays flat (never reaches 0) and the integral diverges. The right tail can be completed, but the estimate of the mean then depends on which completion is used.

### 1.2 First solution: Efron's completion

Setting $\hat{S}(t) = 0$ for $t > t_{\max}$ (Efron) gives $\hat{\mu} = \int_0^{t_{\max}} \hat{S}(t)\,\mathrm{d}t$. Downsides: it assumes all survivors at $t_{\max}$ die immediately afterwards (biased downward), and $t_{\max}$ varies from sample to sample.

### 1.3 Second solution: restricted mean

**Theorem 7.1: Restricted mean**

For a pre-specified $\tau$, estimate $\mu_\tau = \mathrm{E}[\min(X, \tau)] = \int_0^\tau S(t)\,\mathrm{d}t$ by

$$\hat{\mu}_\tau = \int_0^\tau \hat{S}(t)\,\mathrm{d}t, \qquad \hat{V}(\hat{\mu}_\tau) = \sum_{j=1}^{D} \Big[\int_{t_j}^\tau \hat{S}(t)\,\mathrm{d}t\Big]^2 \frac{d_j}{Y_j(Y_j - d_j)},$$

with $100(1-\alpha)\%$ CI $\hat{\mu}_\tau \pm Z_{1-\alpha/2}\sqrt{\hat{V}(\hat{\mu}_\tau)}$. (The sum runs over death times $t_j \le \tau$.)

**Remark 7.2: Why the weights $\int_{t_j}^{\tau} \hat{S}$**

A perturbation of the hazard estimate at $t_j$ rescales $\hat{S}(t)$ for every $t \geq t_j$, hence changes the area to the right of $t_j$ only. By the delta method, the contribution of $\hat{h}_j$ to $\mathrm{Var}(\hat{\mu}_\tau)$ is (area right of $t_j)^2 \times$ Greenwood's term $d_j/[Y_j(Y_j - d_j)]$.

Since $\hat{S}$ is a step function, $\hat{\mu}_\tau$ is a sum of rectangles: $\hat{\mu}_\tau = \sum_k \hat{S}(s_k)(s_{k+1} - s_k)$ over consecutive jump points $s_k$ in $[0, \tau]$.

**Example 7.3: 6-MP group (data of Lecture 5)**

KM: $\hat{S} = 0.857, 0.807, 0.753, 0.690, 0.627, 0.538, 0.448$ at $t_j = 6, 7, 10, 13, 16, 22, 23$ ($Y_j = 21, 17, 15, 12, 11, 7, 6$). With $\tau = 35$ (the largest, censored, time):

$$\begin{aligned} \hat{\mu}_{35} &= 6(1) + 1(0.857) + 3(0.807) + 3(0.753) + 3(0.690) \\ &\quad + 6(0.627) + 1(0.538) + 12(0.448) \approx 23.29 \text{ weeks}, \\ \hat{V}(\hat{\mu}_{35}) &\approx 7.99, \quad \text{SE} \approx 2.83, \quad 95\%\ \text{CI} \approx (17.75,\ 28.83). \end{aligned}$$

## 2 Quantiles and Their Confidence Intervals

Recall $x_p = \inf\{t : S(t) \leq 1 - p\}$ ($p = 0.5$: median). The estimator is $\hat{x}_p = \inf\{t : \hat{S}(t) \leq 1 - p\}$; its variance depends on the density $f(x_p)$, which is hard to estimate. Instead, CIs are obtained by **inverting** the pointwise CIs for $S(t)$.

**Theorem 7.4: CIs for $x_p$**

A $100(1-\alpha)\%$ CI for $x_p$ is the set of all $t$ satisfying

$$\text{linear:} \quad -Z_{1-\alpha/2} \leq \frac{\hat{S}(t) - (1-p)}{\hat{V}^{1/2}[\hat{S}(t)]} \leq Z_{1-\alpha/2},$$

$$\text{log-log:} \quad \left| \frac{[\log\{-\log \hat{S}(t)\} - \log\{-\log(1-p)\}]\,\hat{S}(t) \log \hat{S}(t)}{\hat{V}^{1/2}[\hat{S}(t)]} \right| \leq Z_{1-\alpha/2},$$

$$\text{arcsine:} \quad \left| \frac{2\{\arcsin\sqrt{\hat{S}(t)} - \arcsin\sqrt{1-p}\}\{\hat{S}(t)(1-\hat{S}(t))\}^{1/2}}{\hat{V}^{1/2}[\hat{S}(t)]} \right| \leq Z_{1-\alpha/2}.$$

**Derivation: where these come from**

Each is a Wald test of $H_0 : S(t) = 1 - p$ on the transformed scale: $|g(\hat{S}) - g(1-p)|/(|g'(\hat{S})|\,\mathrm{SE}(\hat{S})) \leq Z$. For log-log, $1/|g'(\hat{S})| = |\hat{S} \log \hat{S}|$; for arcsine-square-root, $1/|g'(\hat{S})| = 2\sqrt{\hat{S}(1-\hat{S})}$. The CI collects all $t$ at which $H_0$ is not rejected.

**Example 7.5: Median for the 6-MP group**

$\hat{S}(22) = 0.538 > 0.5 \geq \hat{S}(23) = 0.448$, so $\hat{x}_{0.5} = 23$. Checking the criteria at each death time, all three methods are satisfied from $t = 13$ onwards ($\hat{S}(13) = 0.690$, SE 0.107) but not at $t \leq 10$. Since $\hat{S}$ never drops far enough below 0.5 to violate the criterion again, the 95% CI

for the median is $[13, \infty)$: the upper limit cannot be determined.

## 3 Right-Censored and Left-Truncated Data

**Definition 7.6: Data and modified risk set**

Data $\{(L_i, T_i, \delta_i)\}$: $L_i$ = age at entering the study, $T_i$ = death or censoring time. At death times $t_1 < \cdots < t_D$ let $d_j$ = number of deaths and

$$Y_j = \sum_{i=1}^{n} I(L_i \le t_j \le T_i),$$

the number who have *entered* and are still under observation at $t_j$.

With this modified $Y_j$, all estimators of K&M Sections 4.2–4.4 (KM, Nelson–Aalen, Greenwood, CIs, bands) apply unchanged.

**Remark 7.7: What is being estimated**

Subjects are only seen after their entry time, so the estimators target *conditional* quantities:

$$S_L(t) = P(X > t \mid X > L), \qquad H_L(t) = -\ln P(X > t) + \ln P(X > L), \qquad H_L'(t) = h(t).$$

The hazard itself is unaffected by conditioning, which is why $d_j/Y_j$ still estimates it.

**Example 7.8: Channing House**

Residents of the Channing House retirement center (California): 97 males and 365 females; truncation times are the ages (months) at entry. The number at risk is small at young ages, rises to a peak around 950–1000 months, then falls.

- Males: the first subject entered at 751 and died at 777; the second entered at 759 and died at 781; the third appeared only at 782.
- Plain KM conditional on the youngest entry: at 777, $Y = 2$, $d = 1 \Rightarrow$ factor $1/2$; at 781, $Y = 1$, $d = 1 \Rightarrow$ factor 0. So $\hat{S}_L(t) = 0$ for all $t \ge 781$ — the estimate is destroyed by the tiny early risk set.
- Remedy: estimate $S_a(t) = P(X > t \mid X \ge a)$ for a chosen age $a$ where the risk set is reasonably large:

$$\hat{S}_a(t) = \prod_{a \le t_j \le t} \Big(1 - \frac{d_j}{Y_j}\Big), \quad \hat{V}[\hat{S}_a(t)] = \hat{S}_a(t)^2 \sum_{a \le t_j \le t} \frac{d_j}{Y_j(Y_j - d_j)},$$

$$\hat{H}_a(t) = \sum_{a \le t_j \le t} \frac{d_j}{Y_j}, \qquad t \ge a.$$

## 4 Left-Censored Data: Reversing the Time Scale

$X$ = failure time, $C$ = left censoring time; observe $T = \max(X, C)$, $\delta = I(T = X) = I(X \ge C)$. How to estimate $S(u) = P(X > u)$?

**Theorem 7.9: Reversed time-scale trick**

Pick $\tau > \max_i T_i$. Then $\tau - X_i$ is *right* censored by $\tau - C_i$:

$$\tau - T_i = \min(\tau - X_i,\ \tau - C_i), \qquad \delta_i = I(\tau - X_i \le \tau - C_i).$$

Let $G(u) = P(\tau - X > u)$. Apply the usual KM to $\{(\tau - T_i, \delta_i)\}$ to get $\hat{G}$, and set

$$\hat{S}(u) = 1 - \hat{G}(\tau - u), \qquad \mathrm{Var}[\hat{S}(u)] = \mathrm{Var}[\hat{G}(\tau - u)].$$

CIs and bands are built the same way.

**Why.** $\max(X, C) = \tau - \min(\tau - X, \tau - C)$, and $X \ge C \iff \tau - X \le \tau - C$. Also $G(\tau - u) = P(\tau - X > \tau - u) = P(X < u)$, so $S(u) = 1 - G(\tau - u)$ (for continuous $X$). ■

## 5 Right-Truncated Data

We observe $\{(R_i, X_i)\}$ sampled conditionally on $X_i < R_i$. Reverse the time scale with $\tau > \max_i R_i$: then $\tau - X_i$ is *left truncated* by $\tau - R_i$ (since $X_i < R_i \iff \tau - X_i > \tau - R_i$). Use the standard left-truncation method to estimate the conditional distribution of $\tau - X_i$ given $\tau - X_i > a$.

**Definition 7.10: Choice $a = \tau - \tau^*$, $\tau^* = \max_i R_i$**

$$G^*(t) = P(\tau - X_i \ge t \mid \tau - X_i \ge \tau - \tau^*) = P(\tau - X_i \ge t \mid X_i \le \tau^*),$$
$$S^*(t) = P(X_i > t \mid X_i \le \tau^*) = 1 - G^*(\tau - t).$$

**Theorem 7.11: KM on the reversed scale**

With $\tau - t_j$ the $D$ distinct failure times on the reversed scale,

$$\hat{G}^*(t) = \prod_{\tau - \tau^* \le \tau - t_j \le t} \left(1 - \frac{d_j}{Y_j}\right),$$

$$d_j = \#\{i : \tau - X_i = \tau - t_j\}, \quad Y_j = \#\{i : \tau - X_i \ge \tau - t_j \ge \tau - R_i\},$$

and $\hat{S}^*(t) = 1 - \hat{G}^*(\tau - t)$. Inference for $S^*$ uses the properties of $\hat{G}^*$ for LT data.

Only the conditional survival given $X \le \tau^*$ is estimable: nothing in the data speaks about $X > \tau^*$.

**Remark 7.12: Quick review**

KM and NA were developed for a single, possibly right-censored failure time with independent censoring. They extend directly to LT + RC data (modified risk set), and to left-censored or right-truncated data by reversing the time scale.

## 6 Other Sampling Schemes

Interval censoring, double censoring, and their combinations with left truncation have no simple closed-form estimator; Turnbull's iterative methods for these cases are covered in Lecture 8.